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In-vivo 3D zebrafish imaging for longitudinal observation of the retina regeneration with different light intensity irradiations

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ABSTRACT

The retina regeneration process has been observed in several animals including fish, birds, and amphibians, whereas the injured human retina cannot regenerate until today. The zebrafish is one of the animals which can regenerate their retina. Due to similarities between humans and the zebrafish in the morphology of the eye as well as the gene, the zebrafish has been chosen as a useful model for investigating retina regeneration. Previous studies have observed morphological changes in the zebrafish retina using optical coherence tomography during the retinal regeneration after light irradiation with a single beam power. However, these studies have limitations of demonstrating differences in regenerative abilities depending on degrees of retinal damages because they used only a single light irradiating condition. Through light-induced retinal injuries by various light irradiating conditions, investigation correlations between the time required for retina regeneration and grades of retinal damages were needed. To conduct these experiments, a custom-built OCT, which can acquire a cross-sectional and three-dimensional image of the zebrafish's eye, was developed based on OpticStudio® simulations. The zebrafish's eye imaging can provide the ability to observe the damage and morphological changes after laser irradiation in the eye tissues including the cornea and retina.

Keywords: Zebrafish, retina regeneration, laser irradiation, optical coherence tomography, OCT

1. INTRODUCTION

Spontaneous retina regeneration of human beings still remains unsolved issues. Humans cannot restore vision impairments of humans from retinal disease including glaucoma and age-related macular degeneration (AMD). On the other hand, the zebrafish can spontaneously regenerate the retina [1, 2]. According to previous studies, the zebrafish's retina regeneration and vision restoration were capable by Müller glial cell-derived progenitors [1, 2]. When retinal damage occurs, zebrafish's resident Müller glial cell reenter the cell cycle and produce proliferative multipotent progenitor cells. These responses lead to the production of new photoreceptors and vision restoration. Another reason that the zebrafish was adopted for retina regeneration studies was the similarities between humans and the zebrafish in the morphology of the eye as well as the gene [3, 4]. In the case of in-vivo investigations, previous studies have observed morphological changes in the zebrafish retina using optical coherence tomography during the retinal regeneration after light irradiation with a single beam power [5, 6]. However, these studies have limitations of demonstrating differences in regenerative abilities depending on degrees of retinal damages because they used only a single light irradiating condition. Studies improving these limitations had to investigate correlations between the time required for retina regeneration and grades of retinal damages by various light irradiations. In addition, these investigations ought to include observations of morphological changes in whole eyes. Although the cornea, lens, and retina were correlated and essential in the whole vision system, previous studies considered only the retina without considering changes in the anterior segment [5, 6]. Two-dimensional (2D) and three-dimensional (3D) structure images of zebrafish's eyes can provide quantitative comparisons of morphological changes in anterior and posterior segments of the zebrafish's eyes during the retina regeneration process. We designed a custom-built OCT system for 2D and 3D imaging of the cornea and the retina of zebrafish eyes through optical design software (OpticStudio®) based on ray tracing.

2. METHODS AND MATERIALS

2.1 Imaging system

In order to acquire in-vivo images of zebrafish eyes, we developed optical coherence tomography (OCT) which is a powerful 3D imaging system providing in-vivo high-resolution volume images. Lightsource of the developed OCT has broad bandwidth (100nm) centered at 840nm (M-D-840HP SUPERLUM, Carrigtwohill, Ireland), which provides approximately 3 μ m theoretical axial resolution. A custom spectrometer of the OCT is composed of a diffraction grating (1200L/mm at 840nm, Wasatch Photonics, Inc. Durham, USA), a focusing lens of the spectrum (Planar T*1.4/85, Carl Zeiss AG, Oberkochen, Germany), and a line scan camera (spL4096-140 km, Basler AG, Ahrensburg, Germany). To build OCT sample arms for zebrafish eyes imaging, an optical design software, OpticStudio® (Zemax, LLC, Seattle, USA), was used to design lens combinations and simulate focal points at the anterior and posterior chamber of the zebrafish.

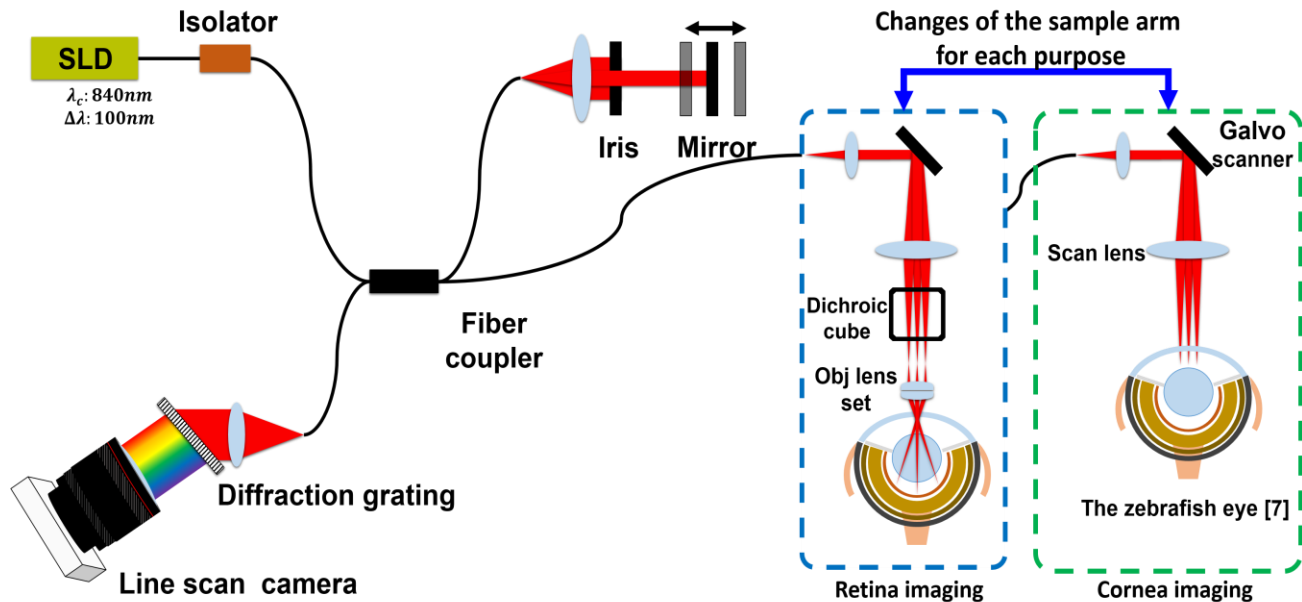


Figure 1. Schematic of the imaging system (custom-built OCT) [7-10]

2.2 The optical design of the zebrafish eye model

Before designing OCT sample arms for anterior and posterior segments, zebrafish eyes were mimicked using OpticStudio®. Previous researches provided specific anatomical values of the zebrafish's eye[11, 12]. We presented these parameters and additional values used to design zebrafish in table 1.

Table 1. Used geometrical value of the zebrafish eyes to make zebrafish eye model

unit: mm

Parameter	Value	Parameter	Value	Parameter	Value
Corneal diameter	2.000	Distance between lens and retina	0.440	Aperture of the iris	1.020
Distance between cornea and lens	0.017	Diameter of the retina	2.215	Diameter of the lens	0.960
Distance between cornea and iris	0.348	Radius of the cornea	2.800	Retina radius	1.107
Zebrafish eye diameter	2.130	Corneal thickness	0.020		

2.3 The optical design of sample arms

Sample arms of the OCT for imaging anterior and posterior segments were designed based on commercial aspherical lens combinations. The sample arm for anterior chamber imaging consisted of only a scan lens between the collimator and target as shown in Fig.2(a). On the other hand, an objective lens set (two aspherical lenses) was applied for posterior imaging to irradiate a smaller beam than the diameter of the zebrafish's pupil and focus on the retina plane. The scanning range of each virtual sample arm was $3131\mu\text{m}$ (for anterior imaging) and $239\mu\text{m}$ (for posterior imaging) in simulations.

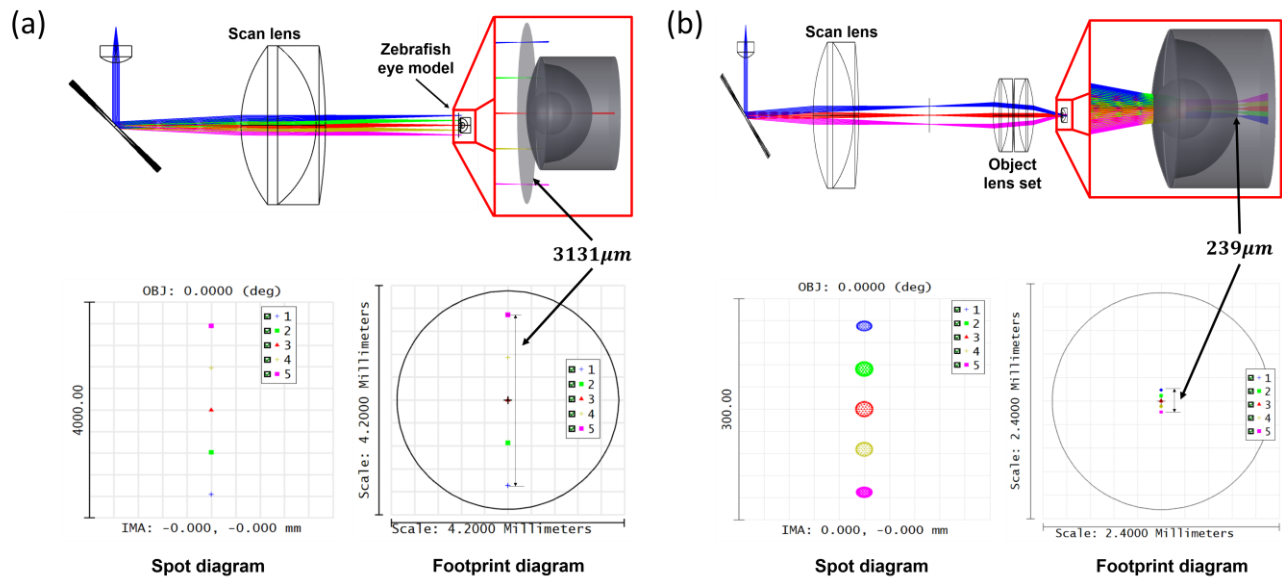


Figure 2. Design and simulation of each sample arm for imaging anterior (a) and posterior (b) segments

2.4 Imaging protocol

The tricaine mesylate solution was used for anesthesia of in-vivo zebrafish imaging referenced as the previous researches and ZFIN (an online biological database of information about the zebrafish) [13]. The zebrafish were posed roughly on wipes in petri-dish after immersing it in anesthesia solution to prevent the movement. Anesthesia states and breathing were maintained by poured anesthesia solution in the petri-dish. Fine posing was performed using multi-axis and tilt linear stages to set a maximally orthogonal position with a center A-scan beam. All imaging processes complied with animal protocol approval by the IACUC of Inha University (Approval number: 200901-717).

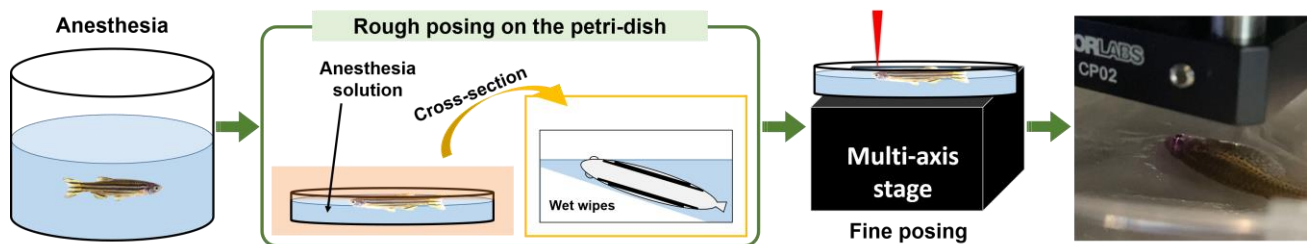


Figure 3. Imaging protocol of a zebrafish study

3. RESULTS

3.1 Anterior segments imaging

We acquired cross-sectional images and volume images from the cornea to the iris of the healthy adult zebrafish (approximately 8 months post fertilization) eye. The cross-sectional image in Fig. 4(a) was a B-scan image nearby at the center of the eye extracted from volume scanning.

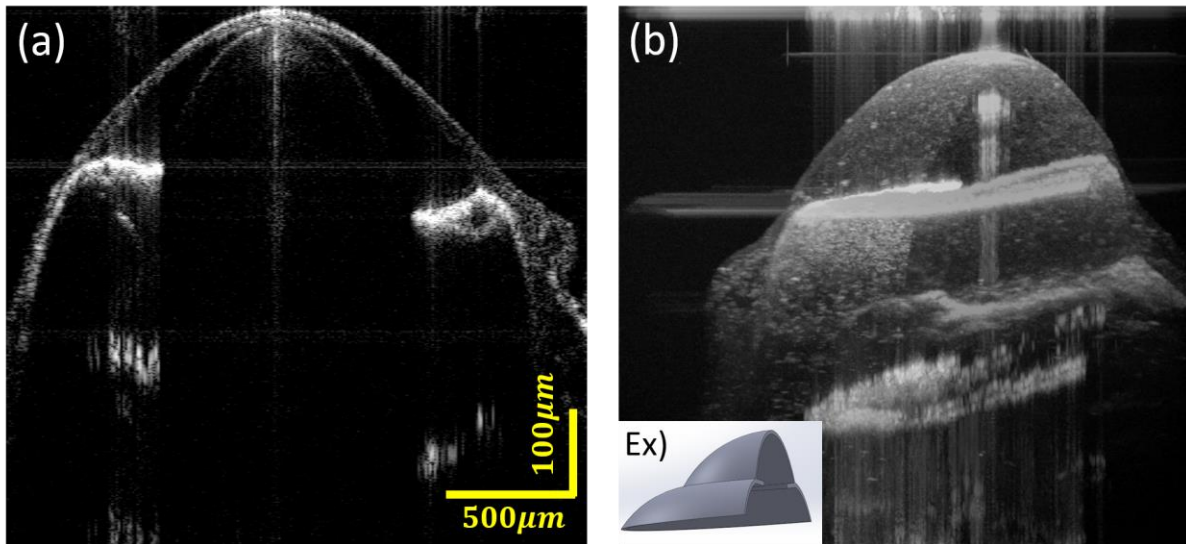


Figure 4. Images of anterior segments of the zebrafish eyes. (a) B-scan image, (b) cross-sectional view of the 3D volume. An example of cross-section 3D volume-rendered by SOLIDWORKS® 2020

3.2 Retina imaging

Retinal images of zebrafish eyes were acquired through a customized sample arm for posterior imaging. Although automatic segmentation algorithms cannot segment the definite boundary of retinal layers, manual segmentation was possible.

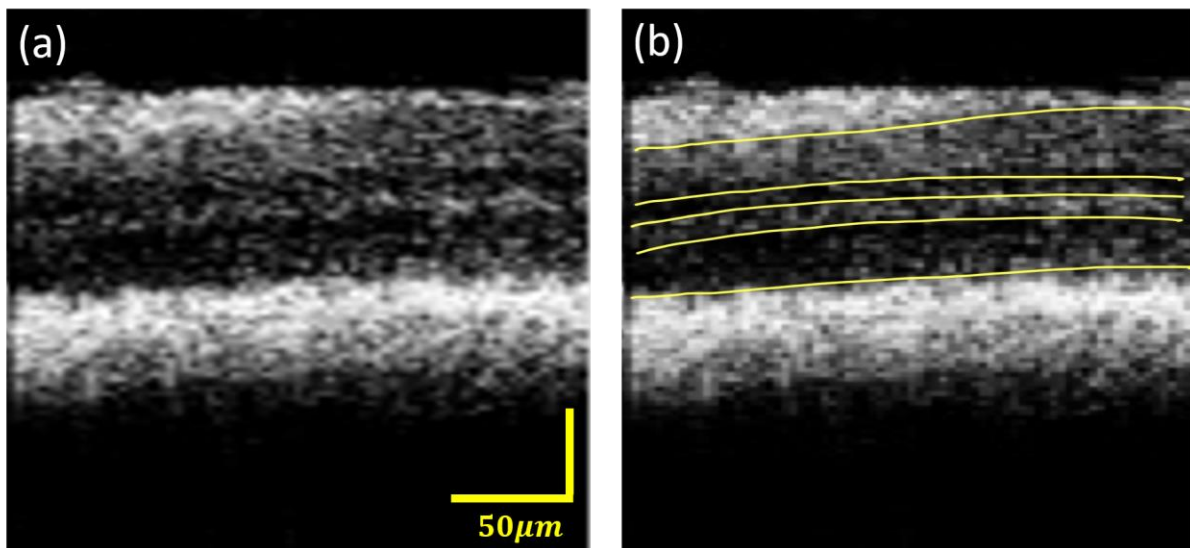


Figure 5. Images of the zebrafish retina. (a) B-scan image, (b) manually segmented image of the (a)

4. CONCLUSION

In this research, we acquired cross-sectional and 3D volume images of the zebrafish eyes using custom-built optical coherence tomography based on design simulated using OpticStudio®. The system with a sample arm for the anterior segment provided high-resolution images including the cornea and iris. These high-resolution images could provide various parameters which can quantify and model morphologies of the zebrafish's anterior chamber. Quantitative anatomical parameters including central corneal thickness, corneal curvature, anterior chamber depth, and anterior chamber angle will be significant factors tracing morphological changes during longitudinal zebrafish monitoring. On the other hand, the resolution of retina images was insufficient to trace slight changes in retinal layer thickness compared to the commercial OCT. We will improve the resolution in retina imaging through optical simulation with a more optimized zebrafish eye model and objective lens set. In addition, laser irradiation with different protocols and longitudinal monitoring observing the spontaneous retina regeneration process will be performed.

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We performed dispersion compensation of OCT imaging through our post-processing software as well as optical alignment of the custom-built OCT system assisted by Dr. Robert J. Zawadzki in UC Davis.

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